

# Student Screening Guidelines

## Selection & Next Steps Process

1. **Initial Submission (Step 1 & Step 2).** Applicants submit an experimental replication summary, a slide deck or report, academic transcripts, and a 5–6 page research proposal.
2. **One-on-One Technical Interview.** Based on the depth of the submission, selected applicants are invited to a 1-on-1 call to discuss their experimental results, research goals, and project feasibility.
3. **Voluntary Research Internship.** Outstanding interview candidates are invited to join the group as voluntary research interns, with direct mentorship targeting top-tier AI/vision/interdisciplinary conferences and journals (e.g., MICCAI, ISBI, CVPR, ICCV, NeurIPS, IEEE TMI/MIA, Nature-family journals).
4. **Ph.D. Position Consideration.** Highly qualified applicants are strongly encouraged to apply for formal graduate admission at Texas Tech University. Admission and funding decisions are competitive and subject to university and departmental requirements; strong performance during the voluntary internship provides substantial internal support toward funded MS or Ph.D. positions for Spring 2027, Fall 2027, or later terms.

## Two-Step Evaluation Tasks

### Step 1: Codebase Exploration & Experimental Verification

Select at least one of our recent papers (LoMix, EMCAD, MK-UNet, EfficientMedNeXt, EffiDec3D, CASCADE, G-CASCADE, or PP-SAM), review its methodology, and run experiments using the corresponding open-source codebase.

- **Deliverable 1:** A brief experimental execution summary with setup notes, run logs, and verified benchmark metrics.
- **Deliverable 2:** A concise technical slide deck (5–7 slides) or a 2–3 page report covering the paper's core contributions, observed bottlenecks, and replication findings.

### Step 2: Research Proposal (5–6 Pages)

Develop a 5–6 page research proposal under one of the following tracks. Brain, cardiac, and general neuroimaging applications are preferred, but there is no strict restriction on imaging modality (e.g., CT, MRI, cardiac imaging, histopathology, endoscopy, ultrasound, retinal fundus, OCT, dermoscopy) or on specific clinical or AI for Good applications.

#### Path A — Lab Paper Extension or Methodological Enhancement

Propose a project that extends our existing codebases beyond segmentation into predictive medicine and digital diagnostics, broader AI for Good challenge, or that directly improves our core methodologies (e.g., architectural efficiency, optimization, or accuracy).

#### Path B — Open Research Proposal Across Core Interdependent Fronts & AI for Good

Propose a project aligned with one of our interconnected research pillars, addressing computational medicine or a broader AI for Good challenge (e.g., healthcare equity, low-resource clinical screening, smart agriculture, climate and environmental monitoring, disaster response, transportation safety, or assistive robotics):

1. **Continuous Adaptation, Security & Reliability.** Test-time adaptation (TTA), prompt-based tuning, federated learning, and robust security frameworks for handling scanner, sensor, environmental, or device shifts. Security and reliability work should address the end-to-end pipeline: automated detection of model vulnerabilities, adversarial attacks, or hallucination in MLLMs/VLMs; precise spatial, feature, or textual localization of threats and errors; and active defense, mitigation, or hallucination-purification strategies across healthcare and AI for Good deployments.

2. **Scalable, Explainable & Interactive AI.** Translating robust, secure backbones into lightweight, high-throughput engines for resource-constrained edge hardware (e.g., mobile devices in rural clinics, point-of-care units, agricultural sensors, onboard robotic systems), with visual attention maps and interactive human-in-the-loop expert feedback.
3. **Generative AI & Comparative Synthesis.** Using interpretable, secure representations to build privacy-aware generative models for few-shot rare-disease synthesis, cross-species or cross-modality diagnostic translation, and synthetic data generation for underserved or low-resource domains.
4. **Multimodal Foundation Models, Reasoning Segmentation & Omics.** Integrating adapted vision encoders and generative representations with genomic, transcriptomic, clinical EHR, geospatial, or sensor data using MLLMs/VLMs for spatial biomarker discovery, complex reasoning segmentation (e.g., segmenting anatomical or pathological structures from free-form clinical queries or multi-step rationale), environmental scene deduction, and automated reporting.

### **Path C — Benchmark & Challenge Ownership**

Identify an active or upcoming competition at MICCAI, ISBI, or a premier AI for Good benchmark challenge (e.g., CVPR/ICCV workshops, Kaggle, Earth observation, disaster response, or smart agriculture benchmarks). Pitch a concrete methodological strategy and an execution plan to take primary responsibility for experimentation and submission under faculty supervision.

### **Required Proposal Structure (5–6 Pages)**

- Motivation and clinical, societal, or methodological significance
- Problem definition and gaps in related work
- Solution direction, proposed methods, and framework diagrams/figures
- Expected outcomes and preliminary results